Cerebellum (2013) 12:632–644
DOI 10.1007/s12311-013-0470-2

ORIGINAL PAPER

Multidimensional Analysis of Fetal Posterior Fossa in Health and Disease

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Published online: 4 April 2013

Abstract Fetal magnetic resonance imaging (MRI) is now routinely used to further investigate cerebellar malformations detected with ultrasound. However, the lack of 2D and 3D biometrics in the current literature hinders the detailed characterisation and classification of cerebellar anomalies. The main objectives of this fetal neuroimaging study were to provide normal posterior fossa growth trajectories during the second and third trimesters of pregnancy via semi-automatic segmentation of reconstructed fetal brain MR images and to assess common cerebellar malformations in comparison with the reference data. Using a 1.5-T MRI scanner, 143 MR images were obtained from 79 normal control and 53 fetuses with posterior fossa abnormalities that were grouped according to the severity of diagnosis on visual MRI inspections. All quantifications were performed on volumetric datasets, and supplemental outcome information was collected from the surviving infants. Normal growth trajectories of total brain, cerebellar, vermis, pons and fourth ventricle volumes showed significant correlations with 2D measurements and increased in second-order polynomial trends across gestation (Pearson r, $p<0.05$). Comparison of normal controls to five abnormal cerebellum subgroups depicted significant alterations in volumes that could not be detected exclusively with 2D analysis (MANCOVA, $p<0.05$). There were 15 terminations of pregnancy, 8 neonatal deaths, and a spectrum of genetic and neurodevelopmental outcomes in the assessed 24 children with cerebellar abnormalities. The given posterior fossa biometrics enhance the delineation of normal and abnormal cerebellar phenotypes on fetal MRI and confirm the advantages of utilizing advanced neuroimaging tools in clinical fetal research.

Keywords Posterior fossa · Cerebellum · Malformation · Fetal MRI · Volumetry · Neurodevelopmental outcome

Electronic supplementary material The online version of this article (doi:10.1007/s12311-013-0470-2) contains supplementary material, which is available to authorized users.

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Introduction

Posterior fossa malformations are one of the main brain anomalies detected during fetal nervous system development [1]. The protracted period of growth, extending from the early thickening of the alar plate around the sixth week of gestational age (GA) to complete maturation in the first postnatal year, increases the vulnerability of the cerebellum to both congenital and acquired pathologies [2–4]. Several radiological phenotypes were replicated by transgenic animal models which have identified numerous genes and transcription factors responsible for normal cerebellar development, and the mutations that are associated with prevalent malformations such as cerebellar/vermis hypoplasia [5–7].

The emerging functional evidence has challenged the confined role of the cerebellum as the centre for control and coordination of motor skills by supporting its modulatory impact on higher cognitive functions such as language, learning and emotions [8]. Recent neurodevelopmental outcome research in children has associated cerebellar malformations with an array of functional impairments, ranging from gross motor disabilities to expressive language problems, some of which were consistent with attention deficit hyperactivity disorder and autism spectrum disorders [9–11]. While the cerebellar vermis involvement was correlated with functional disabilities [12], additional brainstem abnormalities worsened the prognosis for children with malformed cerebellums [13].

Consequently, routine examination of the posterior fossa with ultrasonography (US) is performed between 18 and 22 weeks of gestation as part of an anomaly scan. However, the variability in morphometric analysis impedes a diagnostic consensus in the classification of cerebellar malformations. No objective quantification regarding the relative development of posterior fossa structures has been conducted to confirm the visual diagnosis in either subtle cases of inferior vermis hypoplasia or in more severe cerebellar malformations involving the neocerebellum (Fig. 1a, b). In addition, findings such as mega cisterna magna have been deemed normal variants with no adequate measurements and comparisons with functional outcome [14] (Fig. 1c). Thus, the lack of biometric analysis not only hampers the characterisation of associated malformations but also leads to anecdotal prognostic information provided for parental counselling [15]. This is of paramount importance given the reported termination rates of 80 % in some countries [16].

The complementary role of fetal magnetic resonance imaging (MRI) in evaluating the US suspected abnormalities with higher spatial resolution and tissue contrast is now well documented, and concerns about a cerebellar malformation is one of the main referral reasons for clinical MR examination [1, 17]. However, despite the recurrent use of diametral MRI measurements (e.g. transcerebellar diameter and vermis dimensions) [18, 19], the convoluted structure of the cerebellum requires volumetric analysis to improve the developmental descriptions. Modern advances in imaging sequences and post-acquisition processing techniques now allow for fetal MR data to be utilized for 3D measurements by overcoming the detrimental effects of fetal and maternal motions [20, 21]. The proposed methods can play a crucial role in improving the characteristics of normal and abnormal posterior fossa development.

In addition to the earlier 3D sonographic volumetry of the fetal cerebellum with multiplanar and virtual organ computer-aided analysis methods [22–25], a few MRI studies have reported their findings for normative cerebellar growth over varying GA ranges using image reconstruction and tissue segmentation techniques [26–28]. However, to our knowledge, no study has described the relative development of posterior fossa structures on motion-corrected volumetric images and applied these data to examine cerebellar anomalies in utero. Using reconstructed image datasets, our specific aims were to ascertain the biometrics of normal fetal posterior fossa development across the full range of gestation and to analyse high-risk fetuses with cerebellar malformations in reference to the normative data. We hypothesized that the detailed 3D analysis would enhance the developmental characterisation of posterior fossa structures including neocerebellum, vermis, pons and fourth ventricle. In addition, the identification of developmental abnormalities not detected exclusively by 2D analysis would complement the conventional examination's efficacy in clinical diagnosis and the phenotypic description of cerebellar malformations with their potential influence on the whole brain development. Furthermore, the supplementary information on neurodevelopmental outcome would generate valuable insights on the relationship between the structure and function of the cerebellum, thereby improving prognostic information.

Materials and Methods

The ethical approval for this prospective cross-sectional study was granted by the Hammersmith Hospital Ethics Committee allowing us to obtain research consent for the fetal scans of 512 pregnant patients, performed at the Robert Steiner MRI Unit between June 2007 and June 2011.

Subject Selection and Clinical Classifications

The reasons for referral in the 109 control mothers included previously abnormal pregnancies, US suspected anomalies or healthy volunteers. Subsequent findings of delivery complications (n=1), preterm birth (n=1), positive prenatal infection/genetic/cardiac screenings (n=10), abnormal neonatal MRI/clinical examination/1–2 years neurodevelopmental assessments (n=11), twin pregnancy or poor image quality (n=7) constituted the exclusion criteria resulting in a normal control cohort of 88 scans from 79 fetuses, 40 males and 39 females, with a mean GA of 29.76 weeks (range, 21.71–38.86 weeks; SD, 4.58), determined via the early US examination. The mean maternal age at scan was 30.73 years (range, 17.17–45.51 years; SD, 6.90).

A cohort of 70 fetuses with a confirmed diagnosis of posterior fossa abnormality was selected from the fetal imaging database between January 2008 and April 2011. While 17 fetuses were excluded because of non-reconstructable images (n=8), twin pregnancies (n=7) and posterior fossa crowding (n=2), 53 pregnant women (55 scans) were enrolled with a mean GA of 27.28 weeks (range, 18.57–37.43 weeks; SD, 4.88). The sex of the fetuses was recorded as 30 males and 23 females, and the mean maternal age was 31.21 years (range, 20.93–44.17 years; SD, 5.99).

Apart from the visual description of an *enlarged fourth ventricle*, the cerebellar abnormality and mega cisterna magna classifications complied with the diagnostic criteria outlined in Bolduc et al. [12] to provide terminological consistency with the current literature. A total of five subgroups were created according to the presence of concomitant brain anomalies (Table 1). The distant positioning of the mid-sagittal vermis away from the brainstem separated

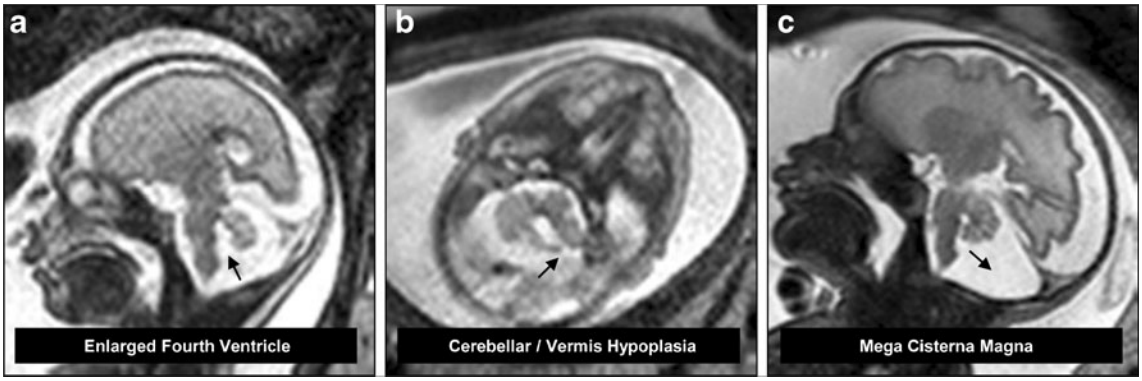


Fig. 1 T2-weighted MR images of **a** Enlarged fourth ventricle (widening of the CSF space with normal cerebellar morphology at 23 weeks of GA, also referred to as inferior vermis hypoplasia), **b** Cerebellar/vermis hypoplasia (reduced cerebellar diametric measurements at 26 weeks of GA), **c** Mega cisterna magna (dilated cisterna magna with normal cerebellar morphology and biometry at 32 weeks of GA)

by an elongated fluid space with normal cerebellar/vermis 2D measurements was termed enlarged fourth ventricle (EFV) (Fig. 1a); the same finding with an additional non-cerebellar anomaly was defined as enlarged fourth ventricle with a supratentorial abnormality (EFVS). This radiological phenotype, EFV, has been referred to as vermis rotation, inferior vermis hypoplasia or Dandy–Walker continuum in previous studies. The cerebellar abnormality group (CA) consisted of apparent cerebellar/vermis hypoplasias (Fig. 1b) or rhombencephalosynapsis with altered morphology and no other brainstem or cerebral anomalies. The fourth group included fetuses demonstrating cerebellar/vermis hypoplasias and additional brainstem and/or supratentorial malformations (CAS) such as pontine hypoplasia, ventriculomegaly, agenesis of the corpus callosum, subependymal heterotopia and polymicrogyria. A final group of isolated mega cisterna magna (MCM) fetuses all had retrocerebellar space greater than 10 mm with normal 2D cerebellar measurements and no other supratentorial abnormality on visual analysis (Fig 1c). The GA was normally distributed in each subgroup.

Neuroimaging, Reconstruction and Quantification Techniques

The pregnant women were scanned in a left lateral tilt position to avoid vena caval obstruction from the pregnant uterus, using a 1.5-T Phillips Achieva scanner with either a 5- or 32-channel cardiac array coil. No sedation was administered, and the mothers were free breathing throughout. A full clinical examination was carried out in axial, sagittal and coronal planes with T2 and T1 weighted, and SNAPIR [29] sequences. For image quantification, datasets were obtained employing a dynamic sequence which oversamples the fetal brain with multiple overlapping T2-weighted single shot slices in three orthogonal planes. Sequence parameters were repetition time=15,000 ms, echo time=160 ms, slice thickness=2.5 mm, slice separation=1.5 mm and flip angle=90 degrees with no refocusing angle. The number of slices for each loop of data varied between 60 and 90 according to the size of the brain.

Post-acquisition processing of raw images was performed on Windows and Linux workstations. The image intensity standardized two sagittal, two coronal and four transverse T2 loops were registered onto a common coordinate space using a robust slice-to-volume registration method which aligned each slice to a precision of 0.3 mm. At the final stage of SVR, the application of multilevel scattered interpolation through MATLAB (version 7.10.0, MathWorks Inc, Natick, MA, USA) produced a 0.74 isotropic reconstruction of true volumetric data with low root mean square error, high signal-to-noise ratio and high-quality contrast [21] (Fig. 2). The volumetric image datasets were then orientated and resampled from a reconstruction voxel size of $1.8\times1.8\times1.8$ mm to $0.2\times0.2\times1$ mm for a detailed analysis.

The 2D measurements were performed on the reformatted transverse and mid-sagittal slices using ImageJ (version 1.45S, National Institutes of Health, Bethesda, MD, USA). The

Table 1 Categorisation of fetal scans illustrating malformations of the posterior fossa

Category	Abbreviation	*n*
Enlarged fourth ventricle	EFV	15
Enlarged fourth ventricle with supratentorial abnormality	EFVS	6
Cerebellar abnormality	CA	9
Cereballar abnormality with supratentorial abnormality	CAS	19
Mega cisterna magna	MCM	6

Abbreviations and sample sizes are outlined for each of the five subgroups

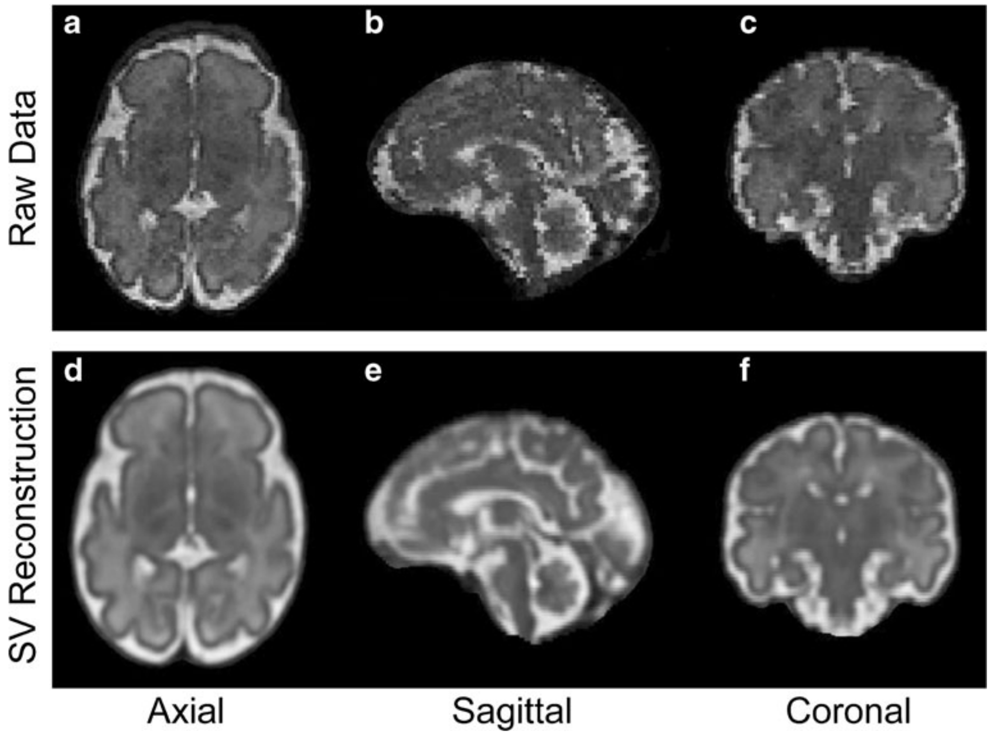


Fig. 2 SVR technique performed on a fetus at 33 weeks of GA. Three orthogonal slices **a**–**c** represent the raw data images acquired from the scanner. Slices **d**–**f** depict a clear comparison of the image quality after the reconstruction process. The 3D volumetric image dataset was produced from eight loops of single shot slices, suitable for linear and volumetric quantifications

standard indices included transcerebellar diameter, quantified as the maximum lateral cerebellar distance on the transverse plane (Fig. 3a 1). The mid-sagittal slice was used to measure the vermis height corresponding to the maximum supero-inferior length (Fig. 3c 3) and the vermis width as the maximum distance between the fastigium and the posterior part of the vermis (Fig. 3c 4). On the same image, the vermis area was calculated by a free-hand drawing tool (Fig. 3b 2), while the tegmento-vermian angle was defined as the degree between a straight line along the dorsal surface of the

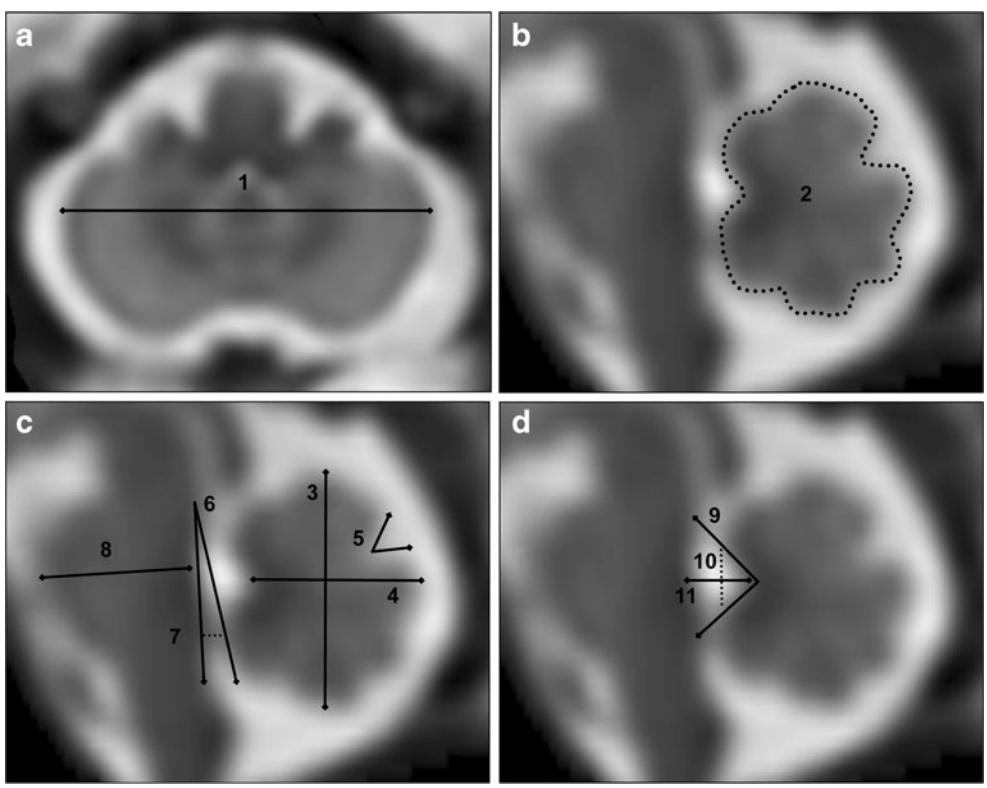


Fig. 3 Anatomical delineations of the 2D measurements performed on volumetric images: **a** transcerebellar diameter (*1*); **b** vermis area (*2*); **c** vermis height (*3*), vermis width (*4*), primary fissure angle (*5*), tegmento-vermian angle (*6*), ponto-cerebellar gap width (*7*), antero-posterior pons width (*8*), **d** fourth ventricle angle (*9*); fourth ventricle height (*10*), fourth ventricle width (*11*)

brainstem from tegmentum until obex and a line along the ventral surface of the vermis (Fig. 3c 6). The pontocerebellar gap was measured from the horizontal distance between the dorsal surface of the brainstem and the most anterior part of the inferior vermis (Fig. 3c 7). The fourth ventricle dimensions were measured as the greatest distance from the posterior part of the pons to the fastigium for the width (Fig. 3d 11); the distance between the superior medullary velum and the obex for the height (Fig. 3d 10); the degree between the superior medullary velum, fastigium and obex for the fourth ventricle angle (Fig. 3d 9); and the degree between the vermian lobules of culmen and declive for the primary fissure angle (Fig. 3c 5). Finally, the antero-posterior pons width was calculated from the maximum distance between the anterior bulge and the ventral surface of the brainstem (Fig. 3c 8).

For 3D measurements, a semi-automatic open source application called ITK-SNAP (version 2.2.0, University of Pennsylvania, Philadelphia, PA, USA) with active contour evolution was utilized [30]. The transverse plane was used for all segmentations because of higher quality, greater contrast and better delineation of anatomy, and the resulting output was cross-checked on two other orthogonal planes (Fig. 4). Image intensity thresholds were established for each region of interest (ROI) decided by the rater. Active bubbles were then placed on strategic locations, and their evolution for covering the ROI was assessed in real time, providing a detailed control of the segmentation method. In the fine editing stage, the rater manually corrected the erroneously segmented voxels. The volume of the desired brain region was calculated by the software, based on multiplying the segmented area with the slice thickness.

The pons volume was measured by segmenting the area defined by the pre-pontine cistern anteriorly, the inferior cerebellar peduncles/dentate nucleus posterolaterally and the fourth ventricle posteromedially. The superior and inferior slice boundaries were confirmed by visualization of the pontine bulge on the mid-sagittal view. The fourth ventricle volume was measured by segmenting the slice starting from the superior medullary velum down until the obex surrounded by pons and the cerebellar vermis. The cerebellar hemispheres were clearly visualized in the transverse plane outlined by the contrast between the CSF and the cerebellar tissue, while the vermis segmentation included the area confined by the two cerebellar hemispheres and the dentate nuclei. The total cerebellar volume measurement included both the cerebellar and the vermis volumes. Furthermore, the additional supratentorial brain quantification excluded brainstem structures below the appearance of the pons, whereas the lateral ventricles quantifications omitted cavum septum pellucidum and vergae.

All 2D and 3D measurements were performed by a single rater (DV) for consistency via pre-determined anatomical delineations based on fetal atlases and the input from two fetal neuroradiologists. The intra-class correlation coefficient for the absolute agreement of both the inter- and intra-observer variability of 3D (n=30) and 2D (n=80) measurements was 0.99, even in difficult delineations such as vermis volume. While for 2D measurements the mean inter-rater difference was 0.12 mm (SD=0.49), with an intra-rater mean difference of −0.0053 mm (SD=0.29), the 3D measurements showed an inter-rater mean difference of 0.034 mL (SD=0.095) and an intra-rater mean difference of −0.0024 mL (SD=0.049). No evidence of systematic error was detected in the Bland–Altman plots (Fig. A.1 in Electronic Supplementary Material), and the average 3D

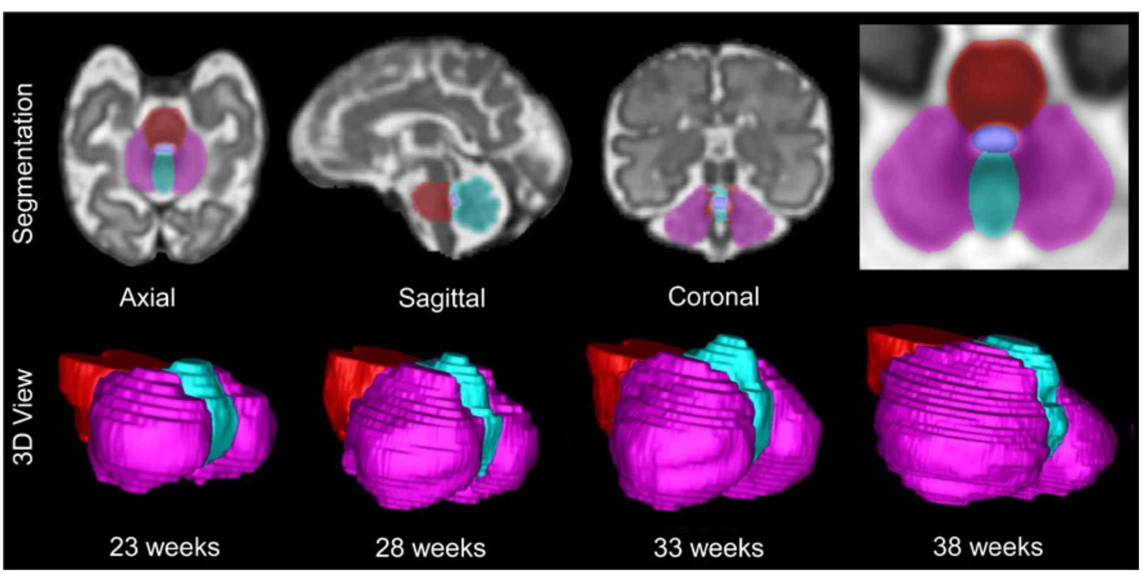


Fig. 4 Semi-automatic segmentation of fetal posterior fossa in three different orthogonal planes and the resulting 3D view across gestation from 23 to 38 weeks in 5-week intervals (sizes do not represent the true scale). The colour coding indicates pons (*red*), cerebellar hemispheres (*purple*), vermis (*green*) and fourth ventricle (*blue*)

measurement Dice coefficient was 0.90 for inter-rater and 0.95 for intra-rater variability, confirming the morphometric similarity of our segmentations.

Postnatal Assessment

Supplemental outcome information on the termination of pregnancy, neonatal death, genetic diagnosis, neonatal and later developmental assessments at ages 1 and 2 years was obtained from 89 % of the 53 abnormal fetuses. Neonatal MRI was offered for clinical investigations, and neurodevelopmental assessments were performed using Griffith Mental Development Scale (Revised) and Bayley Scales of Infant Development, 3rd Edition. No follow-up information could be collected for six fetuses because of missing contact information or the young age of the babies to be assessed, although all were documented as born uneventfully at term. The assessments are ongoing, and further information is being gathered prospectively. The normal control cohort was also assessed with similar methods in order to identify any abnormalities that were not detected prenatally and to ensure normal development.

Statistical Analysis

The statistical analysis was carried out on the SPSS package for Windows (version 19.0, SPSS Inc, Chicago, IL, USA). The data were first assessed for normality using Kolmogorov–Smirnov goodness-of-fit test and Q–Q plots. Logarithmic transformations were performed in order to normalize the quantifications, and simple linear regressions were run on all measurements to find the Pearson correlation coefficients for GA, 2D and 3D measurements. The relative growth rates were calculated for the given GA range. The normal reference centiles agreed with the methods outlined in Royston and Wright [31]. For comparing normal and abnormal cerebellar groups, multi- and univariate ANCOVAs were performed using GA as a covariate and Bonferroni for multiple comparisons correction. The estimated per cent group differences at mean GA for all posterior fossa structures are reported. A confidence level of 0.05 was considered significant.

Results

Normal Reference Data

Biometric indices covering the supratentorial brain, total cerebellar, hemispheric, vermis, pons and fourth ventricle volumes were generated from a normal control cohort of 88 fetal brain images ranging between 21.71 and 38.86 gestational weeks. All 3D measurements had significant correlations with GA, and the best fit models depicted second-order polynomial lines (Fig. 5; Table A.1 in Electronic Supplementary Material).

The relative growth rate of cerebellar volume at 15.5 % per week with a 14-fold increase in mean volume (1,426.3 %) from 1.6 to 22.9 mL across the selected GA range exceeded the supratentorial brain growth at 11.9 % with an 8-fold increase (765.2 %) from 47.1 to 360.1 mL. The vermis-to-cerebellar hemispheric volume ratio depicted a deceleration in vermis growth which was confirmed by the relative growth rates of 16.1 % for the cerebellar hemispheres and 11.9 % for the vermis over the span of 17 weeks (Fig. 6). The growth rates of the pons and fourth ventricle were 9.6 and 12.0 %, respectively.

The 2D measurements also showed strong correlations with GA except for the tegmento-vermian angle ($p=0.272$) (Table B.2 in Electronic Supplementary Material). Most 2D dimensions increased across gestation with best-fit linear models. However, the fourth ventricle and primary fissure angles depicted linear decreases for the given gestational age range, whereas the tegmento-vermian angle failed to illustrate a correlation ($p=0.723$) (Fig. 7).

All 3D measurements showed significant correlations with their respective 2D counterparts significant at the $p=0.001$ level, except for the tegmento-vermian angle ($p=0.117$) (Table C.1–4 in Electronic Supplementary Material).

Cross-Comparison of Cerebellar Anomaly Groups

In comparison to the normal control biometric data, the EFV group ($n=15$) with an abnormal vermis positioning showed no significance on multivariate analysis of 2D measurements except for widening of the fourth ventricle width and height, pontocerebellar gap, tegmento-vermian and primary fissure angles on univariate analysis. However, in addition to a multivariate significance, an estimated 10.6 % decrease in vermis volume and a 197.9 % increase in fourth ventricle volume were reflected in 3D measurements with no evidence of significant differences in supratentorial brain ($p=0.349$), total cerebellar ($p=0.063$), cerebellar hemispheric ($p=0.105$) or pons volumes ($p=0.958$) (Fig. 8).

Compared with normal controls, the addition of a supratentorial abnormality in the EFVS group ($n=6$) was associated with significant differences in all 2D measurements apart from the vermis width ($p=0.372$), fourth ventricle ($p=0.241$) and primary fissure angle ($p=0.639$). The 3D measurements indicated decreases of 25.5 % for total cerebellar, 24.3 % for cerebellar hemispheric, 30.6 % for vermis, 16.2 % for pons and an increase of 232.7 % for fourth ventricle volumes as compared to the normal controls, but there was no difference in the supratentorial brain volume ($p=0.375$) (Fig. 8).

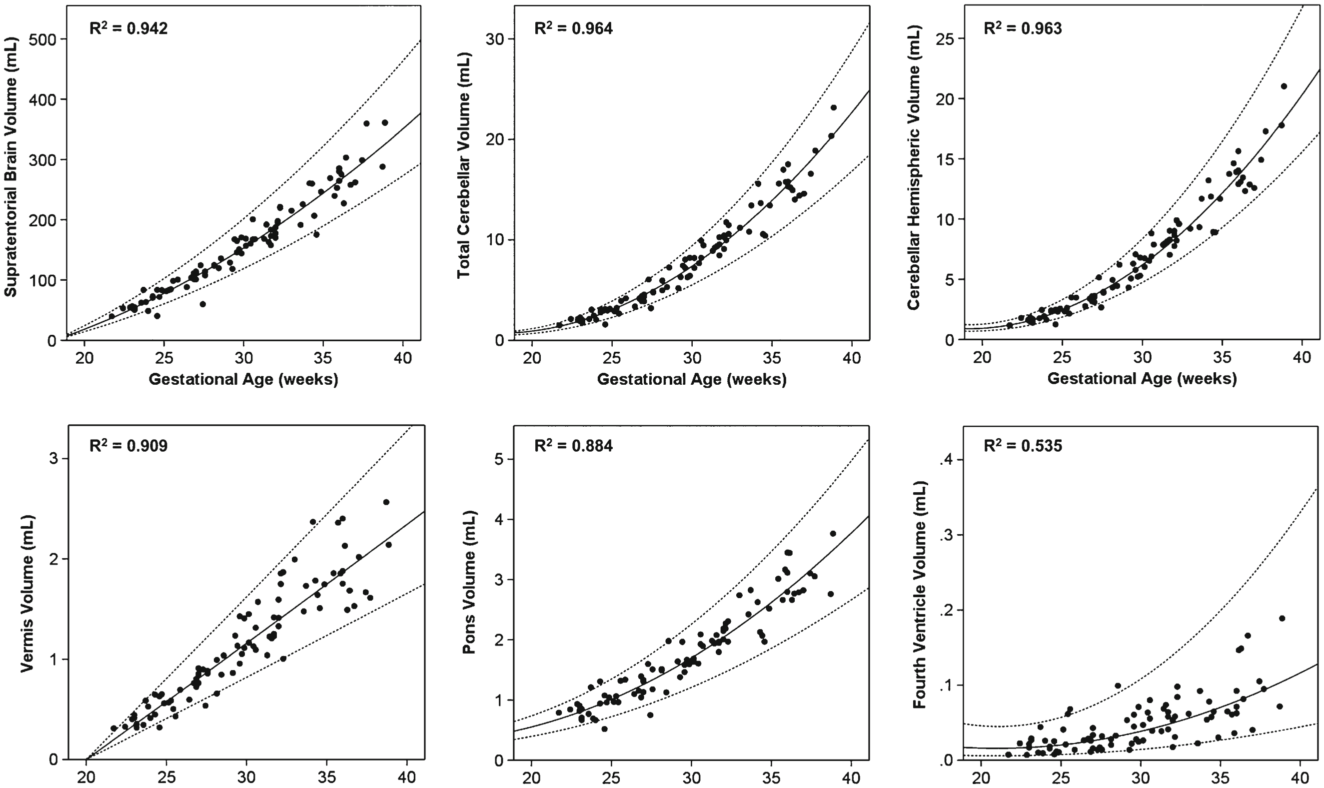
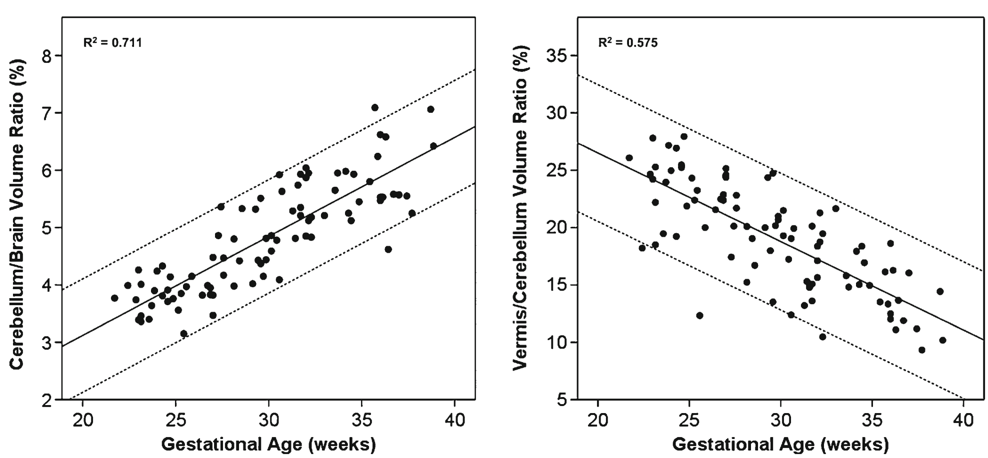


Fig. 5 The best-fit second-order polynomial models ($p<0.001$) for normal control 3D growth trajectories of the supratentorial brain, total cerebellar, hemispheric, vermis, pons and fourth ventricle between 21.71 and 38.86 weeks of GA. The *bold lines* represent the predicted 50th centile values, while the *upper* and *lower dotted lines* correspond to the 95th and 5th centiles, respectively

The cross-comparison between the two enlarged fourth ventricle groups revealed significant univariate decreases in vermis height, pons width and transcerebellar diameter for the subjects with supratentorial abnormalities, which failed to show a group alteration in multivariate analysis (p=0.165). However, a multivariate group difference was apparent in 3D measurements with significant reductions in the total cerebellar (21.0 %), cerebellar hemispheric (20.1 %), vermis (24.5 %) and pons volumes (17.1 %) in the presence of an additional supratentorial abnormality (Fig. 8), yet no difference was detected in the supratentorial brain (p=0.583) and fourth ventricle volumes (p=0.596) on univariate analysis.

As expected in the CA group (n=9) with evident vermis/cerebellar hypoplasia or rhombencephalosynapsis, all 2D measurements were significantly altered when compared to the normal controls except for AP pons width (p=0.365) and fourth ventricle angle (p=0.261). In 3D assessment, there were decreases of 29.4 % for total cerebellar volume, 29.5 % for cerebellar hemispheric volume and 30.6 % for

Fig. 6 The per cent ratios of cerebellum/supratentorial brain and vermis/cerebellar hemispheric volumes across gestation ($p<0.001$). *Bold line* indicates 50th centile, and the *dotted lines* represent 5th and 95th centile boundaries



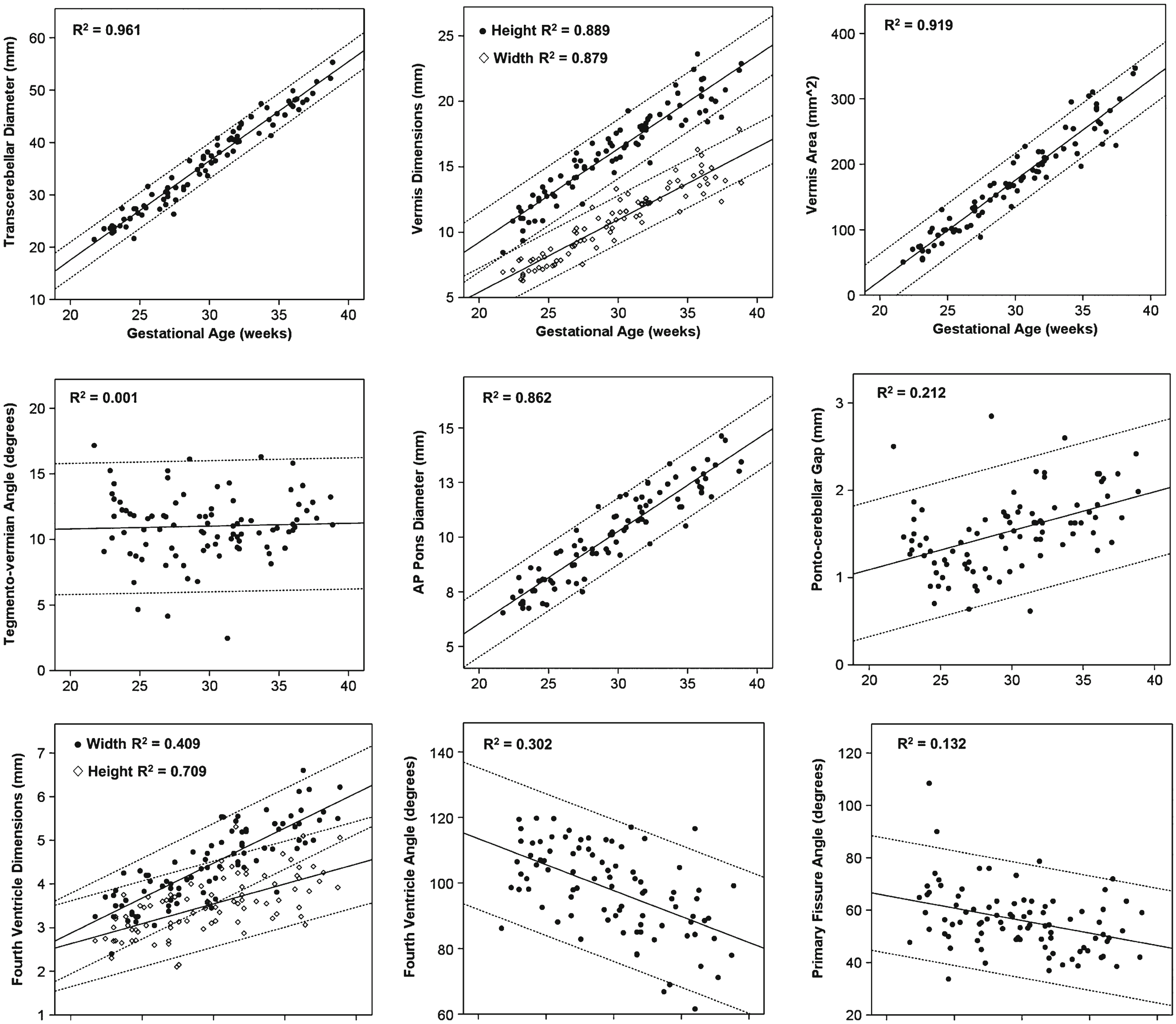


Fig. 7 The best-fit linear models for normal control 2D growth trajectories of transcerebellar diameter, vermis dimensions, vermis area, tegmento-vermian angle, pons diameter, ponto-cerebellar gap, fourth ventricle dimensions and primary fissure angle. All models fit significantly ($p<0.001$) except for tegmento-vermian angle ($p=0.723$). *Bold lines* represent the 50th, while *dotted lines* depict the 5th and 95th centiles

vermis volume and an increase of 160.9 % for fourth ventricle volume, whereas supratentorial brain ($p=0.098$) and pons ($p=0.076$) volumes were not significantly different between the two groups (Fig. 9).

In comparison with normal controls, the CAS group ($n=19$) with additional clinical presentations of pontine hypoplasia, ventriculomegaly, agenesis of the corpus callosum, subependymal heterotopia and polymicrogyria demonstrated smaller transcerebellar diameter, vermis height and vermis area and greater tegmento-vermian angle, pontocerebellar gap and fourth ventricle width. There were also significant decreases in 3D measurements. Lower volumes of supratentorial brain (16.3 %), total cerebellar (25.6 %), cerebellar hemispheric (25.3 %), vermis (31.2 %) and pons (16.2 %) and a higher volume of fourth ventricle (92.2 %) were detected in the abnormal group. Comparing CA to CAS, the multivariate differences did not reach significance in either the 2D ($p=0.907$) or the 3D quantifications ($p=0.359$).

In six fetal scans with isolated mega cisterna magna, there were larger pontocerebellar gap and smaller primary fissure angle and pons width, with no differences in any other 2D quantifications when compared to normal controls. There were no significant differences in 3D measurements, except for an 18.1 % increase of supratentorial brain volume in the MCM group on univariate examination.

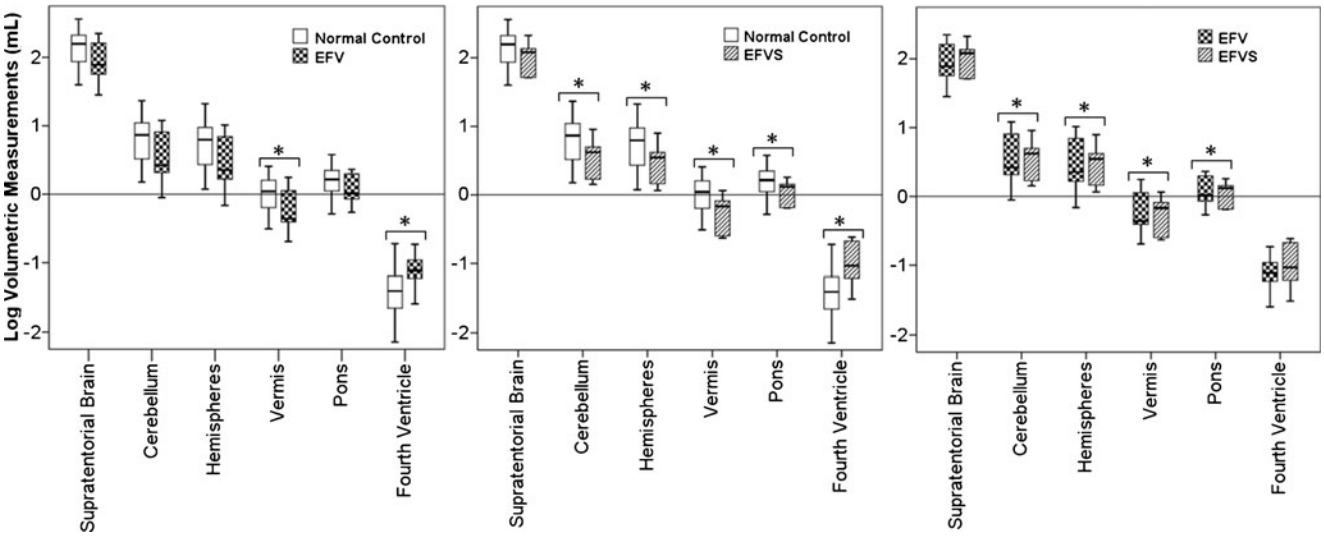


Fig. 8 *Boxplots* representations of 3D quantifications comparing normal controls, enlarged fourth ventricle (*EFV*) and additional supratentorial abnormality (*EFVS*) groups (*$p<0.05$)

Neurodevelopmental Outcome

Out of 53 fetuses with cerebellar abnormalities, there were 15 terminations of pregnancy (28 %) and 8 neonatal deaths (15 %), 5 of which had confirmed genetic abnormalities including Trisomy 18 and 13 and Fryns syndrome. In the remaining 30 surviving infants, a total of 7 genetic diagnoses were made (Trisomy 21, Ohdo syndrome, Type II Waardenburg and Turner syndrome), 13 further infants showed a spectrum of neurodevelopmental delays, 4 were normal and no outcome information could be obtained from 6 infants, all born at term with no clinical concerns. The fetal diagnosis was confirmed with neonatal MRI performed on eight EFV, two CA, two CAS and two MCM babies (100 %). The distribution of available outcome information in the five subgroups is illustrated in Table 2.

The neurodevelopmental delays seen in the EFV group consisted of shortened attention spans, expressive language impairments, visuo-spatial perception difficulties and motor control problems (assessment range=10.0–25.5 months postnatal age). The CA groups depicted pronounced motor delays such as decreased trunk and limb tone and broad based gait (assessment range=12.5–28 months postnatal age). Although the sample size was small, neurodevelopmental delays were also seen in half the MCM group and included visuo-spatial perception and attention problems (assessment range=12.5–25.5 months postnatal age).

Discussion

The lack of objective quantifications and in depth analysis of fetal posterior fossa development has limited our understanding of normal and abnormal growth. With the aim of characterising this complex process in utero, we report on multidimensional assessment of the posterior fossa structures across the full length of pregnancy via advanced neuroimaging tools. Between 21.71 and 38.86 weeks of GA, the supratentorial brain, cerebellar,

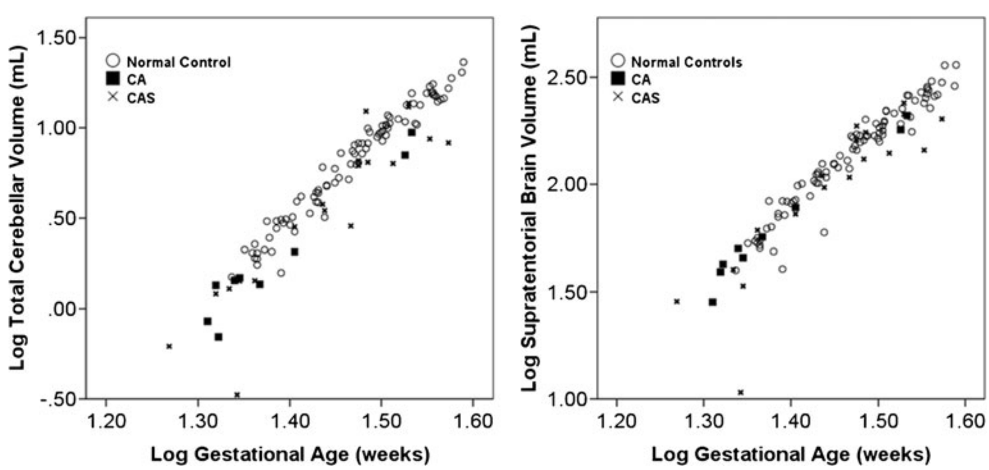


Fig. 9 Scatter plot comparisons of log transformed total cerebellar and supratentorial brain volumes in normal control, CA and CAS groups

Table 2 Distribution of functional outcome information for the cerebellar abnormality groups

Category (*n*)	Alive			Deceased		No developmental information
	Normal	Developmental Delay	Genetic Anomaly	Termination of Pregnancy	Neonatal Death	
EFV (14)	–	6	2	2	–	4
EFVS (6)	–	–	1	1	3	1
CA (9)	1	2	-	4	1	1
CAS (18)	–	2	4	8	4	–
MCM (6)	3	3	–	–	–	–

hemispheric, pons and fourth ventricle volumes showed an accelerated increase, in which the relative growth rate of the cerebellum exceeded that of the cerebrum as illustrated by a recent morphometric analysis [32]. While the vermis to cerebellar volume ratio depicted a decrease in the relative vermis growth rate, pons and fourth ventricle volumes steadily increased in the given GA range. All the 2D measurements increased linearly across gestation and correlated strongly with their 3D counterparts expect for the tegmento-vermian angle.

The comparison of posterior fossa malformations to the control reference data revealed volumetric reductions in the vermis of the EFV group that could not be detected with 2D quantifications or visual analysis. The isolated cerebellar hypoplasias in the CA group did not affect the global supratentorial brain volume. However, the addition of a supratentorial abnormality both in the EFVS and CAS groups was associated with volumetric alterations in all of the posterior fossa structures. Despite the lack of significant differences between the MCM and normal control groups, the supratentorial brain volumes were significantly greater in fetuses with an enlarged cisterna magna. In addition, the neurodevelopmental outcome information confirmed the associated functional impairments in all our subgroups, with only a minority (29 %) of assessed infants showing normal development.

Normal Development and Growth

The reported total cerebellar volumes were in close agreement with the limited data available from previous US and MRI studies, except for deviations in later gestations probably due to the wider age range, robust reconstruction, and detailed tissue segmentation techniques utilised in the present study compared to the methodologies in prior reports [26, 27, 33, 34].

Despite the historical accounts of two distinct cerebellar primordia on the origins of cerebellar formation, conventional views now suggest the presence of an inverted V-shaped single primordium at the dorsolateral part of the alar plate acting as the basis of cerebellar growth, where the zipper-like closure of the primordial limbs form the cerebellar vermis [7]. During this event, the maturation of vermis precedes that of the phylogenetically newer neocerebellum [35], which is also evident by the decreasing rate of vermis growth in our given GA range. On the other hand, the close coupling of the accelerated cerebellar and pontine growth rates strengthens the embryological evidence of a common origin of progenitor cells for the development of these two structures. In conjunction with the documented dual mesencephalic and metencephalic basis in cerebellar formation, our findings emphasize the importance of analysing the pons and brainstem alongside any cerebellar assessment [36].

With regard to the development of the fourth ventricle, the thinning of the dorsal neural tube surface around 4th week of gestation forms a membranous area that gives rise to the later formation of choroid plexus, the foramen of Magendie and finally to the foramina of Luschka, perforating around 10th and 14–17th weeks, respectively [37, 38]. A limited number of volumetric MRI studies have reported small increases in the lateral ventricle dimensions across gestation [39, 40]; however, a significant change in the fourth ventricle volume has not been shown. The accelerated increase in the fourth ventricle volume in our study might be explained by the cerebrospinal fluid dynamics and the morphological changes in the convexity and concavity of the surrounding vermis and cerebellar hemispheres.

Even though the main emphasis of this article was placed on 3D quantifications, all 2D cerebellar measurements also showed significant linear relationships with fetal gestation except for tegmento-vermian angle which would be expected to stay stable for a morphologically normal cerebellar anatomy and caudal vermis growth akin to the brainstem [41]. The conventional measures of transcerebellar diameter and vermis dimensions closely matched the published values [3, 18]. However, despite the significant model fit, a considerable inter-individual variability was observed in the fourth ventricle and primary fissure angles, pontocerebellar gap and fourth ventricle dimensions that can be attributed to the variability in the anatomy and the relatively small delineation of these measurements. Nevertheless, all but tegmento-vermian angle significantly correlated with their volumetric counterparts indicating a close coupling of 2D and 3D measurements in observing normal cerebellar anatomy.

Fetal Posterior Fossa in Cerebellar Anomalies

The isolated enlargement of fourth ventricle on a mid-sagittal MR image is a structural assessment that suffers from inconsistent application of diagnostic criteria, ranging between normal variant to an outdated terminology called Dandy–Walker continuum. On visual inspection, the supposed anticlockwise rotation away from the brainstem indicates either an elevation and/or a hypoplasia of the inferior vermis. While the term "enlarged fourth ventricle" was not intended to introduce a new terminology in the already disordered literature, the variable prognostic data warranted an objective quantification of this structure before determining the presence of a suggested hypoplasia [42].

Comparing normative 3D data to the EFV subgroup illustrated a significant increase in the fourth ventricle volume as expected from the enlarged appearance on the mid-sagittal and transverse slices. While the 2D biometry depicted no differences, we have shown a significant group reduction in vermis volume in association with the visual finding of an enlargement of the fourth ventricle. Although this outcome might suggest that the observed phenotype reflects a degree of vermis hypoplasia, it should be noted that the reduction represents a group difference while the individual data points fell within the lower side of the normal range. Moreover, the areal ratio of the mid-sagittal anterior to posterior vermis lobes did not depict a group difference in comparison with normal controls. The presence of hypoplasia in an abnormally positioned vermis cannot be the sole explanation for cognitive impairment in 23 % of cases in published studies or in our six infants with developmental delays [9]. Further studies analysing the vermis shape and the regional lobular volumes could provide more conclusive results.

The CA group showed significant differences from normal controls in all 3D and 2D measurements except for the total brain and pons. Consistent with our results, a recent study conducted in children with isolated cerebellar malformations depicted no significant changes in the global cerebral volume when compared to normal controls; however, volumetric reductions in the cerebellum were shown to alter the regional cerebral volumes in the central gray matter (i.e. basal ganglia and thalamus) and specific white matter tracts, indicating a need to further optimise advanced techniques to investigate regional structures and connectivity during fetal development [43].

The volumes of the posterior fossa structures illustrated further alterations in EFVS and CAS when compared to the normal controls and the isolated groups. The diagnosis of an additional supratentorial abnormality is commonly associated with more severe neurodevelopmental impairments [44]. The mechanism for this exacerbation is unclear but may reside within the cerebellum and its intrinsic functional connection to the cortical regions involved in higher cognitive functions [45].

Isolated mega cisterna magna is considered a normal anatomical variant with limited follow-up reports suggesting either a normal outcome or subtle neurodevelopmental delays [9] and impairments in adulthood [46]. In a small number of fetuses with isolated MCM, we have demonstrated no differences in 3D measurements as compared to the normal controls except for a significant increase in the total brain volume indicating a possible malfunction in the cellular migration or apoptosis. When compared to the normal controls, the lateral ventricles were also significantly enlarged, but the atrial diameter remained below 10 mm. With respect to the 2D measurements, the MCM group also had a larger vermis width and an increase in primary fissure angle. Previous ultrasound studies have reported ventriculomegaly as the main concomitant finding and have also suggested favourable outcomes [47, 48]; however, the clinical findings of developmental delay in two children of our MCM group and the mechanisms behind these observations are yet to be ascertained.

The limitations of our study include small sample size in each sub-group and the generalized categorisation of the abnormal cases. The GA was added as a covariate in log-transformed group comparisons, but dissimilarities still remained in the median GA between the categories because of the limited sample size; therefore, increased numbers and a stricter classification scheme might be beneficial in future prospective studies. In addition, longer-term follow-up to identify school age cognitive and behavioural impairments is necessary.

Conclusion

Our understanding of normal fetal brain development has long been limited by the lack of biometric data due to poor resolution, the presence of fetal motion and the limited availability of a large control cohort. Such obstacles have restricted our knowledge on deviations from normal growth and have given rise to a classification scheme that cannot be universally accepted. In this study, we provide biometric data for a normal control cohort of 79 fetuses across the GA of 21.71 to 38.86 weeks, utilizing motion-corrected 3D volumetric images and a comparison to five subgroups of cerebellar malformations. We hope that the information provided demonstrates the feasibility and benefits of applying advanced fetal neuroimaging tools to the clinical investigations of fetal growth and that it proves useful for both researchers and clinicians alike. Adding to this research, our future goal is to conduct diffusion tensor imaging to map out the structural connectivity of white matter tracts, regional volumetric analysis, functional connectivity studies and neurodevelopmental follow-up research, all of which will improve characterisation of both normal and abnormal fetal development.

Acknowledgments This study was funded by Biomedical Research Foundation and Medical Research Council, UK. We would like to thank Dr. Ash Ederies, staff at the Robert Steiner MRI Unit, Hammersmith Hospital, London for their support, and all the mothers for their participation in our research.

Conflict of Interest None of the authors have any conflict of interest to disclose.

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